

Summary 1st day

Model building

Objective

To **construct 3D atomic models** of biomolecules (typically proteins) based on experimental or predicted data.

1st step: Map Preprocessing.- enhance map sharpness

- resolution (local deblur)
- connectivity
- trade-off

2nd step: Initial Model

- **Alphafold**

- ▶ Highly accurate, especially for single-chain proteins (5000)
- ▶ No experimental structure is used
- ▶ Main input sequences
- ▶ Relies on coevolution

- **ModelAngelo**

- ▶ Integrates structural constraints (3D maps)
- ▶ Works better at resolution less than 3.5 Å

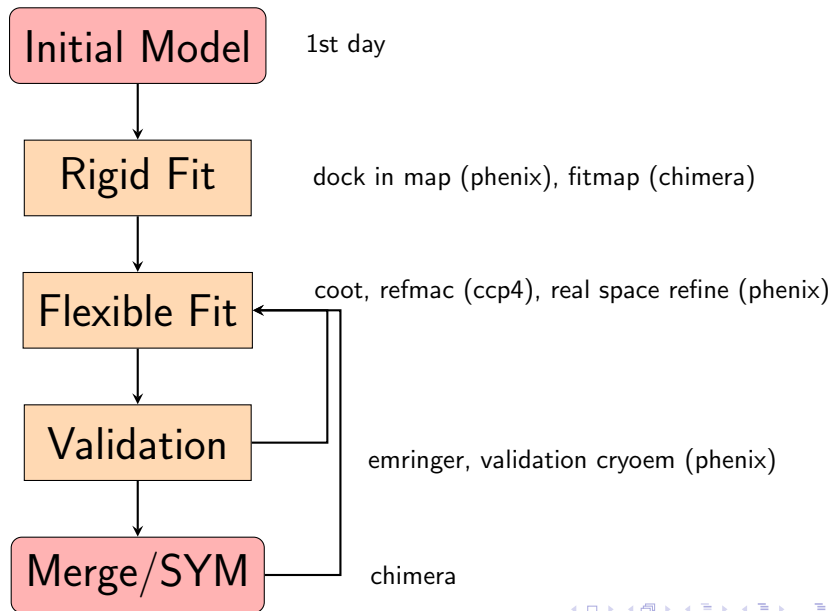
- **Modeller (Sequence Homology Modeling)**

- ▶ Builds models based on known structures of similar sequences
- ▶ Relies on evolutionary conservation
- ▶ Effective when homologous templates are available

Summary 2nd day

Model building

Workflow



Checklist

- Hem group?
- Overlap/clash α -subunit?
- alphafold